



Firmware update and Fallback mechanisms

Index



- Fallback
 - Why Fallback
 - Definition
 - Working
 - Implementation
- Block Diagram
- Memory Layout of STM32F7 Series
- Flash Memory Layout
- Boot-loader Flow

Fallback



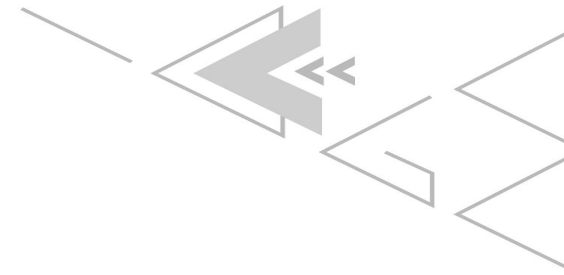
- ▶ Why fallback?
 - ▶ To prevent any system failure due to corrupted, incomplete or malicious firmware or its updates.
 - ▶ This process ensure that the system can recover and still boot up properly.
- ▶ When it is required?
 - ▶ Whenever booting of new firmware result in crash or timeout, which causes interrupt in updating process.
 - ▶ It required to boot the system successfully.

- ▶ How it helps?

Scenario	Without Fallback	With Fallback
Power loss during update	Application area corrupted	Boot up with previous version
CRC failure after update	System hangs or crashes	Boot up with last version firmware
Malicious code update	Runs unsafe code	CRC fails results into recovery mode

- ▶ What if fall back is not implemented?
 - ▶ If any failure happens while system is updating, then system can become non-functional and can't be able to boot up.
 - ▶ Due to failure of the system we have to do the recovery work manually.

Fallback

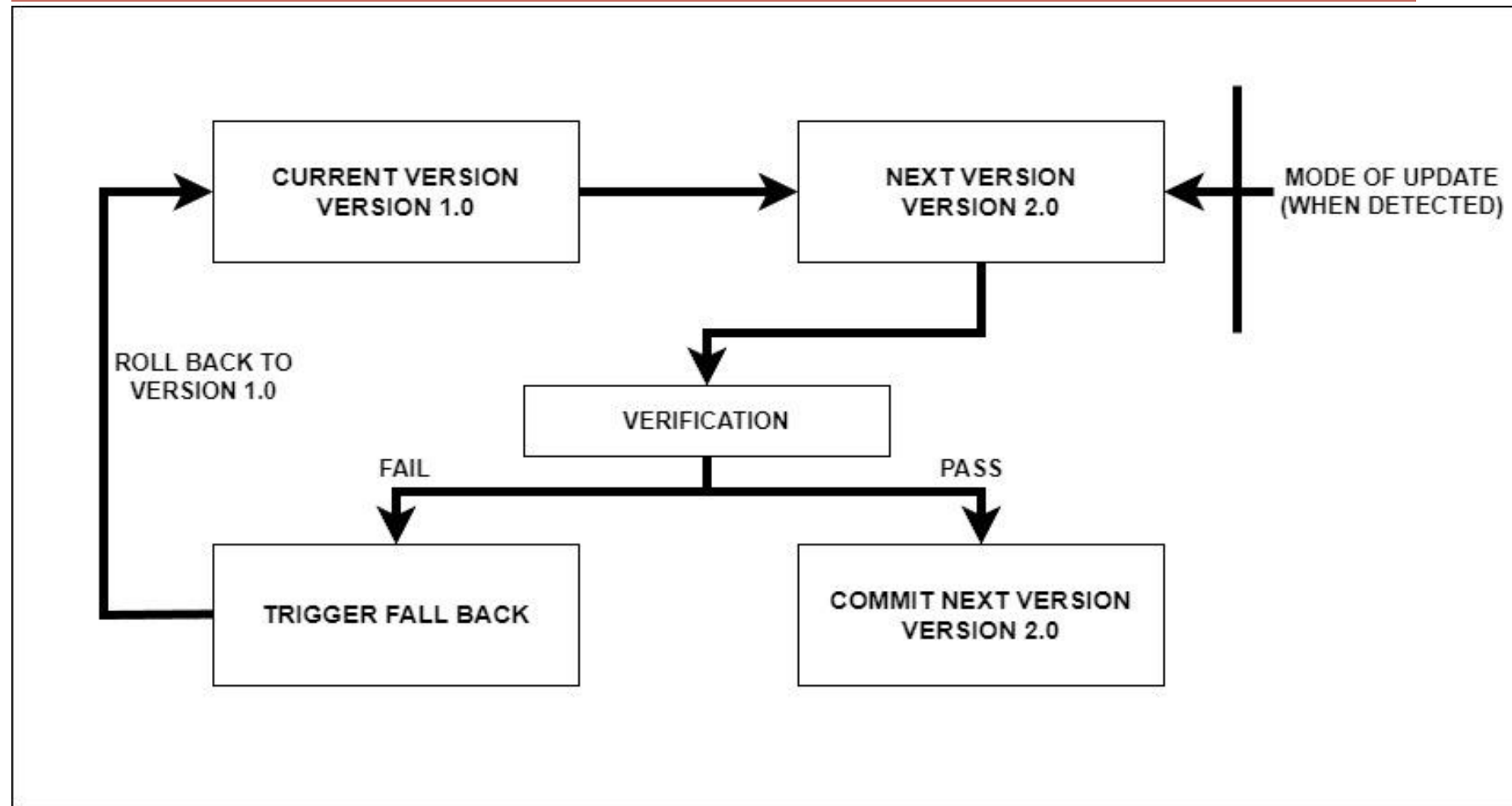


- ▶ Where and how fallback is detected?
 - ▶ Whenever system starts-up the fallback get detected in the boot-loader section.
 - ▶ The boot-loader checks if main firmware is corrupted or if fallback flag is set up properly in the boot-loader.
- ▶ Boot mood selection with help of Boot0 and Boot1

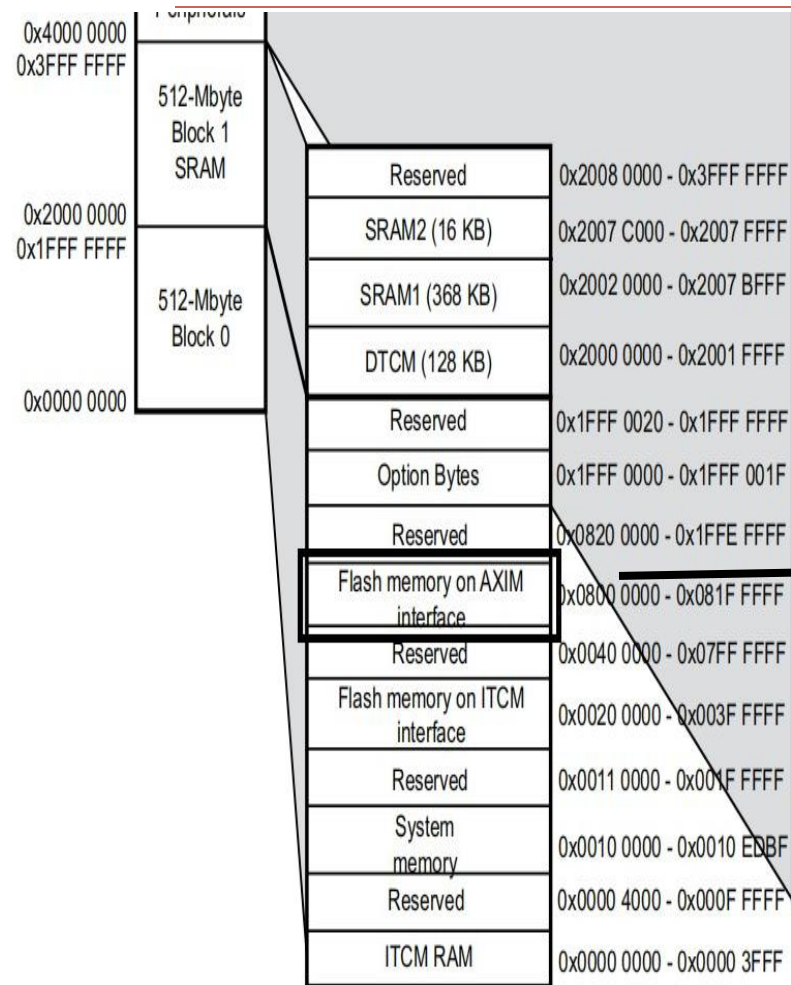
Boot1	Boot0	Boot Mode	Source Target
X (Don't care)	0	Main flash memory	User Application
0	1	System Memory	ST factory boot-loader

- ▶ Memory Utilization
 - ▶ If STM32 flash (2MB) is not enough we have to store the files in external memory.
 - ▶ SDRAM - We can use it to store the temporary files or buffer data.
 - ▶ Flash - We can use the flash to store firmware's logs files, config data and settings files.

> Block Diagram



> Memory Layout of STM32



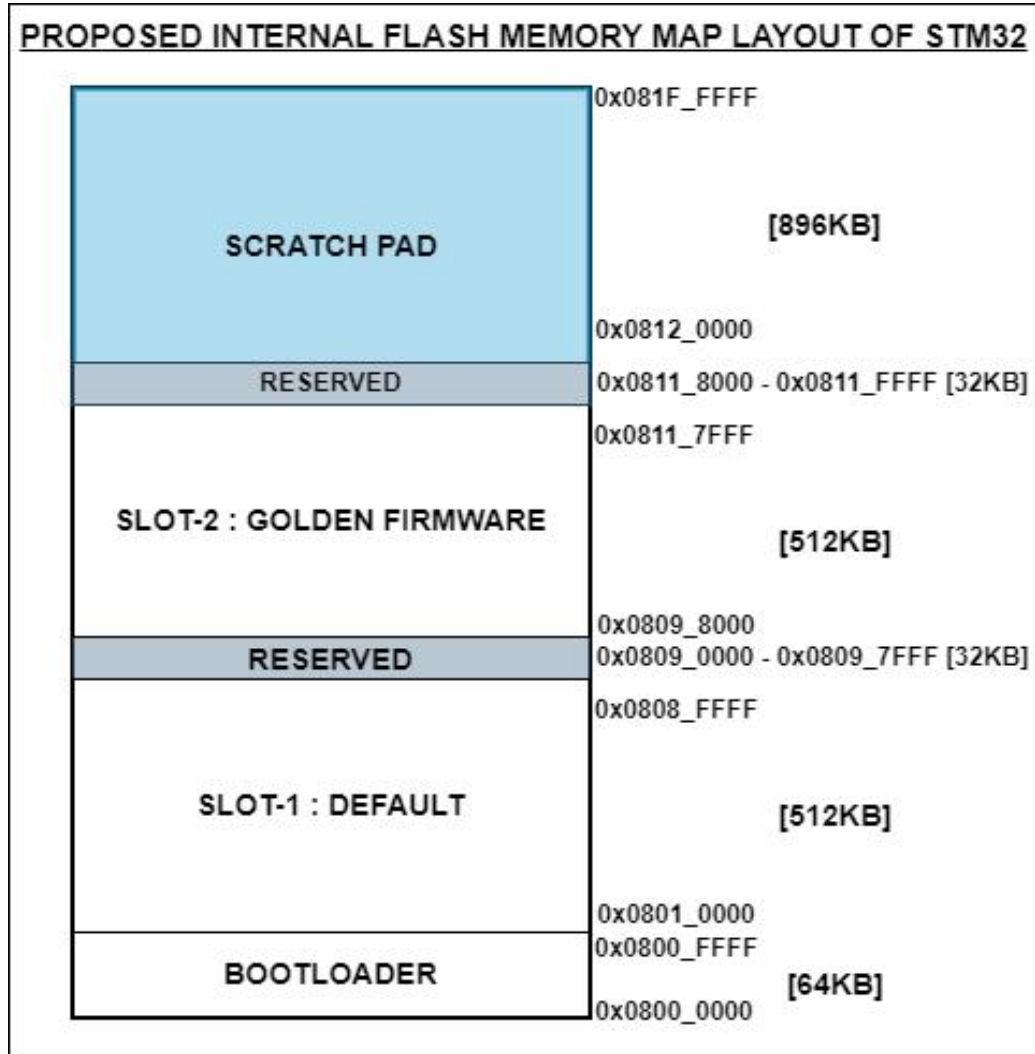
Block	Name	Bloc base address on AXIM interface	Block base address on ITCM interface	Sector size
Main memory block	Sector 0	0x0800 0000 - 0x0800 7FFF	0x0020 0000 - 0x0020 7FFF	32 KB
	Sector 1	0x0800 8000 - 0x0800 FFFF	0x0020 8000 - 0x0020 FFFF	32 KB
	Sector 2	0x0801 0000 - 0x0801 7FFF	0x0021 0000 - 0x0021 7FFF	32 KB
	Sector 3	0x0801 8000 - 0x0801 FFFF	0x0021 8000 - 0x0021 FFFF	32 KB
	Sector 4	0x0802 0000 - 0x0803 FFFF	0x0022 0000 - 0x0023 FFFF	128 KB
	Sector 5	0x0804 0000 - 0x0807 FFFF	0x0024 0000 - 0x0027 FFFF	256 KB
	Sector 6	0x0808 0000 - 0x080B FFFF	0x0028 0000 - 0x002B FFFF	256 KB
	Sector 7	0x080C 0000 - 0x080F FFFF	0x002C 0000 - 0x002F FFFF	256 KB
	Sector 8	0x0810 0000 - 0x0813 FFFF	0x0030 0000 - 0x0033 FFFF	256 KB
	Sector 9	0x0814 0000 - 0x0817 FFFF	0x0034 0000 - 0x0037 FFFF	256 KB
	Sector 10	0x0818 0000 - 0x081B FFFF	0x0038 0000 - 0x003B FFFF	256 KB
	Sector 11	0x081C 0000 - 0x081F FFFF	0x003C 0000 - 0x003F FFFF	256 KB
Information block	System memory	0x1FF0 0000 - 0x1FF0 EDBF	0x0010 0000 - 0x0010 EDBF	60 Kbytes
	OTP	0x1FF0 F000 - 0x1FF0 F41F	0x0010 F000 - 0x0010 F41F	1024 bytes
	Option bytes	0x1FFF 0000 - 0x1FFF 001F	-	32 bytes

Erase flash memory
Erase external memory

Note for sector inde...
Erase selected sect...
Full chip era...

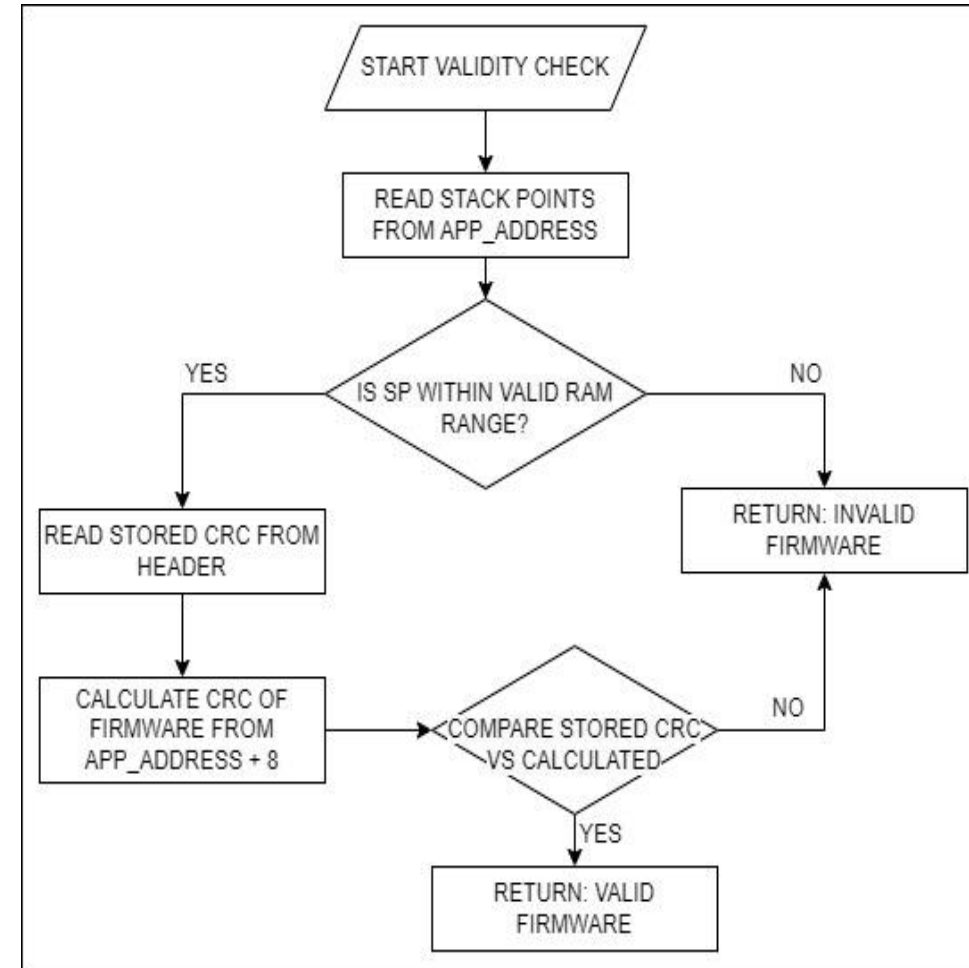
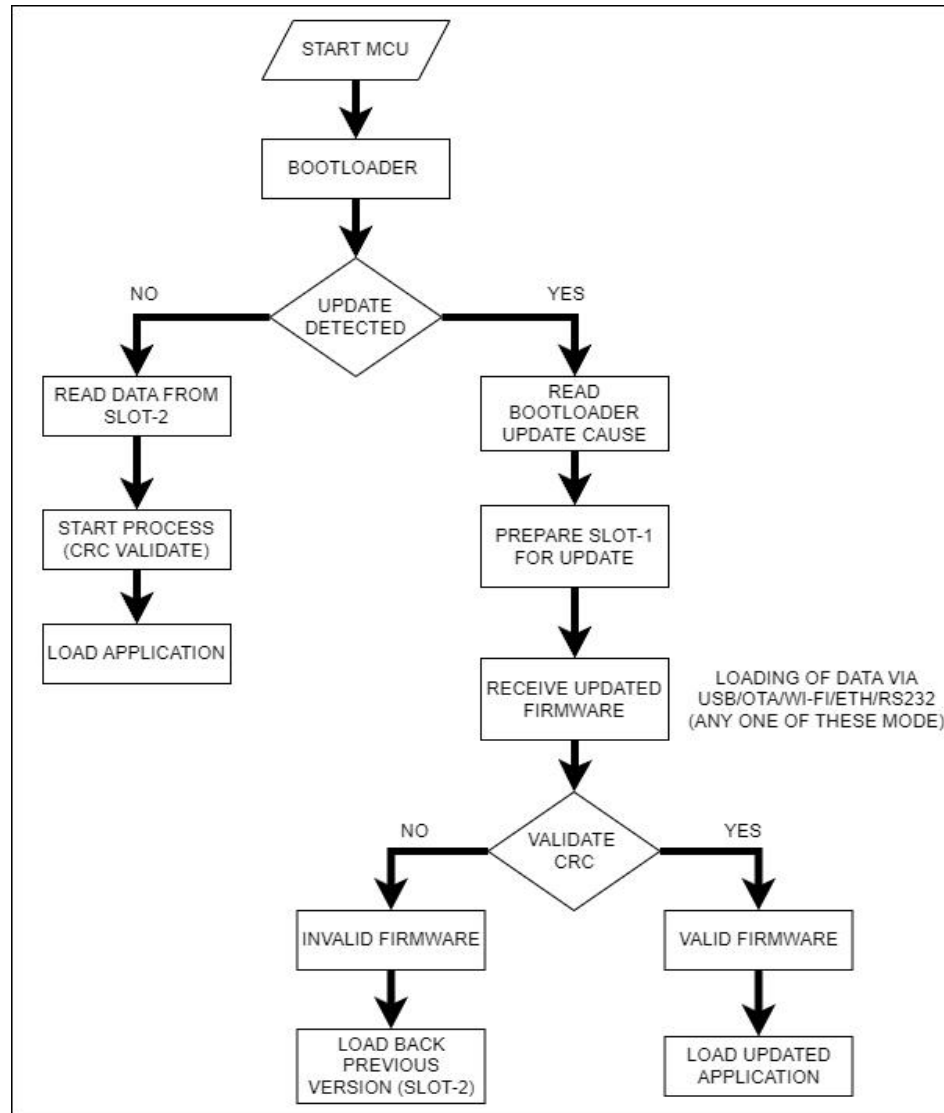
Select	Index	Start Address	Size
☐	0	0x08000000	32K
☐	1	0x08008000	32K
☐	2	0x08010000	32K
☐	3	0x08018000	32K
☐	4	0x08020000	128K
☐	5	0x08040000	256K
☐	6	0x08080000	256K
☐	7	0x080C0000	256K
☐	8	0x08100000	256K
☐	9	0x08140000	256K
☐	10	0x08180000	256K
☐	11	0x081C0000	256K

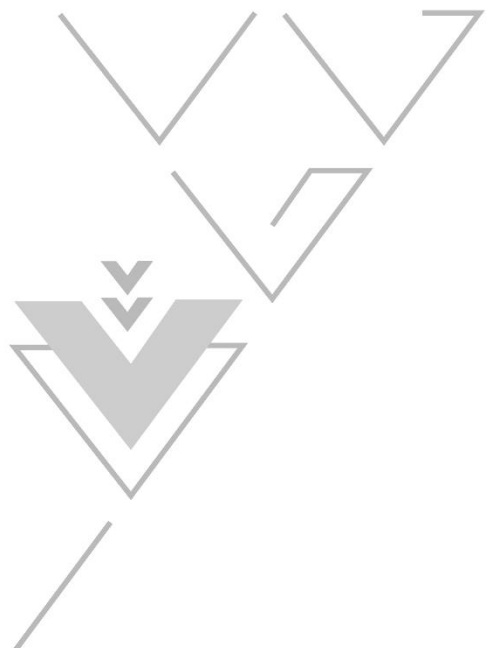
➤ Flash Memory Organization



- ▶ Linker Setup
 - ▶ BOOTLOADER:
FLASH(rx) : ORIGIN : 0x0800_0000, LENGTH = 64K
 - ▶ SLOT-1 : DEFAULT
FLASH(rx) : ORIGIN : 0x0801_0000, LENGTH = 512K
 - ▶ SLOT-2 : GOLDEN FIRMWARE
FLASH(rx) : ORIGIN : 0x0809_8000, LENGTH = 512K
- ▶ Bootloader:
 - ▶ Handles startup, firmware selection and update process.
- ▶ Slot-1 : Default Firmware:
 - ▶ Main firmware; system execution run from here; replaced when new firmware get detected.
- ▶ Slot-2 : Golden Firmware:
 - ▶ Permanent backup firmware; factory version for recovery.

➤ Boot-loader Flow





Thanks!

